Lab11 : Maintenance and Planning

1. Requirements Modeling & Specification

Feature

1. Google Sign-in (Authentication and User Profile)
2. Portfolio Upload (Portfolio Management)
3. User Role Management (Admin Functionality)

Use case diagram

Actors:

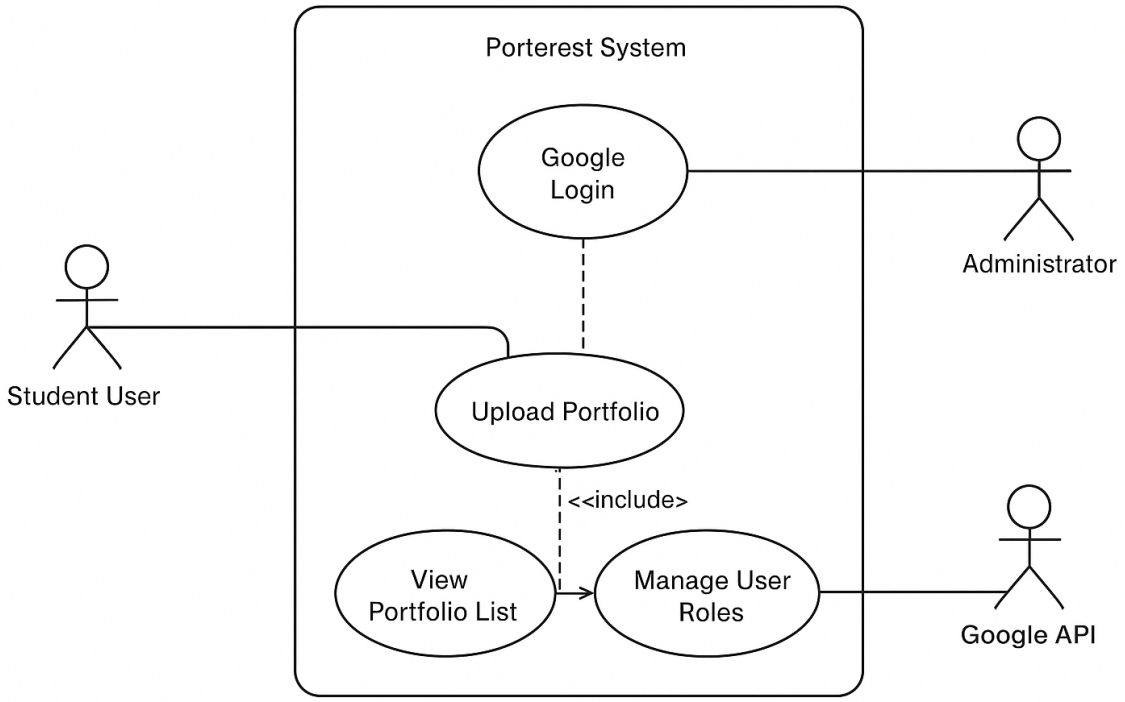
User (Student) – logs in and uploads portfolios

Google API – external authentication provider

System (Porterest) – processes login and manages portfolios

Use case:

1. Sign in using Google account
2. Upload a portfolio
3. View portfolio list
4. Logout



Software Requirement Specification (SRS) Snippets

1. Google Sign-in (Authentication and User Profile)

Functional Requirements	Non-Functional Requirements
- The system must display a “Sign in with Google” button on the login/register page. - When the user clicks the Google sign-in button, the system shall redirect to the Google Auth verification page. - Upon successful verification, the system must create or update the user’s profile in the database with the retrieved information (Full Name, Email, Profile Picture).	- The login process must complete within 3 seconds. - OAuth 2.0 protocol must be used for secure authentication. - Token validation must occur before every session. - Error messages during failed authentication must be clear and instructive.

- The system must automatically log in the user after successful verification. - If the Google account email is not registered, the system must create a new account and request the user to accept Terms & Conditions.	

2. Portfolio Upload (Portfolio Management)

Functional Requirements	Non-Functional Requirements
- Only logged-in users can access the "Upload Portfolio" page. - The user must provide a portfolio title, description, and at least one image. - The system must store portfolio data in the database after submission. - The uploaded portfolio must be publicly visible within 10 seconds. - The system must display confirmation once the upload is successful.	- Upload and save operation must complete within 10 seconds under normal conditions. - Upload, edit, and delete actions must only be available to authenticated users. - The upload page must be intuitive and responsive on desktop and mobile. - Only .jpg, .jpeg, and .png files under 10 MB are accepted.

3. User Role Management

Functional Requirements	Non-Functional Requirements
- The Administrator must be able to view all registered users on the admin dashboard. - The Administrator must be able to change a user's role between "Student," "Viewer," and "Administrator." - The system must update user permissions immediately after the role change. - When a user's account is "Deactivated," they must be unable to log in or access system features. - The system must record every admin action with a timestamp and admin ID.	- The admin dashboard must be secure, requiring administrator authentication for access. - Role or status changes must be logged for audit purposes. - The interface must clearly distinguish active vs. deactivated users. - Role changes should apply across the system within 5 seconds.